Agenthon Track 2: numeric_v1.1

Phase 2.5 • 15 October 2026 • 2,000 draws • Lower scores are better

Decision

Frozen selection: **A1_fhs5**. CV mean improves by **0.21%** versus numeric_v1. The fixed C1 rule and held-out major-group veto determine acceptance, with no added candidates after scoring. Practice proxies participate in selection; they are not independent validation. **The gain is not statistically established: every paired 95% CI includes zero.** Historical caps rise from 23 to 24; the known monthly and rate-trend weaknesses remain unresolved.

Partition	Cards	numeric_v1	numeric_v1.1	Caps v1 → v1.1	v1.1 median
CV 2009–2018	1073	0.953077	0.951081	1 → 1	0.706113
Practice proxies	88	2.435614	2.415744	9 → 9	1.365856
Report-only 2019–2024	554	1.456363	1.454208	17 → 18	0.807978
All historical	2060	1.154244	1.152537	23 → 24	0.759077

Paired year-block bootstrap

Partition	Mean difference	95% percentile CI	Year wins
CV 2009–2018	-0.001996	[-0.005101, +0.000993]	7/10
Practice proxies	-0.019870	[-0.041089, +0.003438]	15/22
Report-only 2019–2024	-0.002155	[-0.010985, +0.004177]	4/6
All historical	-0.001707	[-0.004565, +0.000716]	13/20

10,000 whole-year resamples, seed 20261015, pooled card-weighted means. Negative differences favor v1.1. Intervals include dependence within each year but do not correct for selecting models on these same data. Previously inspected 2019–2024 is report-only except for a fixed >15% group-regression veto.

Fixed-menu ablations

Each stage compares alternatives with the best entering that stage. All candidates are retained, including rejected ones. Full previous-best identifiers, medians, errors, CIs, and breakdowns are in ablations.json and the companion CSVs.

Candidate	CV mean	Caps	Practice mean	Caps	2019–24 mean	Caps	Decision
numeric_v1	0.953077	1	2.435614	9	1.456363	17	Baseline
A1_t5	0.944728	1	2.443867	9	1.451030	17	Reject
A1_mixture	0.944169	1	2.472831	10	1.460843	17	Reject
A1_fhs5	0.951081	1	2.415744	9	1.454208	18	Kept
A2_63	1.057331	0	2.341555	9	1.399092	15	Reject
A2_126	1.034196	0	2.353575	9	1.403075	16	Reject
A2_252	1.006287	0	2.367163	9	1.411043	16	Reject
A3_0.25	0.966306	1	2.410641	9	1.441536	18	Reject
A3_0.5	0.994654	1	2.359592	9	1.426048	16	Reject
A4_damp025	0.949968	1	2.491765	10	1.439492	18	Reject
A4_damp05	0.946507	1	2.479900	10	1.437041	18	Reject
A4_window36	0.951304	1	2.490762	10	1.437440	18	Reject
A4_m0	0.954830	1	2.512595	10	1.448952	19	Reject
A5_pool	0.945439	1	2.453559	9	1.456947	19	Reject
A6_sign01	0.955201	1	2.391356	9	1.475665	18	Reject
A6_sign025	0.976674	0	2.344106	8	1.462074	19	Reject

Acceptance and ordering

C1 requires a lower CV mean, or at most +0.005 worse with fewer combined CV+practice caps; practice mean must not worsen, combined caps cannot rise, and no major CV/practice group may worsen >10%. Major means ≥40 cards, or ≥20 for monthly macro. The reserved-year veto is >15% against either previous best or numeric_v1. Ties within 0.0005 prefer fewer changed knobs. Year groups themselves are descriptive.

A1 isolates tail shape by normalizing variance. A2 uses exp(–h/τ) recent/long-run variance weights. A3 floors variance. A4 changes only monthly handling. A5 pools whole joint draws 50/50; A6 replaces daily drift with momentum-sign drift.

Reserved-year behavior

The precommitted veto checks major family, panel, horizon and single/joint groups. These results did not rank candidate performance or expand the menu.

As-of year	Cards	v1 mean	v1.1 mean	Change	Caps
2019	91	1.3561	1.3540	-0.2%	0 → 0
2020	95	1.4570	1.4624	+0.4%	1 → 1
2021	110	2.0428	2.0500	+0.4%	12 → 12
2022	119	1.7295	1.7288	-0.0%	4 → 5
2023	97	0.7542	0.7341	-2.7%	0 → 0
2024	42	0.9839	0.9773	-0.7%	0 → 0

Group	Cards	v1 mean	v1.1 mean	Change	Caps
factors_daily	77	1.0850	1.0665	-1.7%	1 → 1
g10_fx_daily	183	1.0864	1.0918	+0.5%	2 → 2
g10_fx_daily+rates_daily	20	1.7232	1.7337	+0.6%	1 → 1
macro_monthly	26	2.0143	1.9397	-3.7%	2 → 2
rates_daily	248	1.7646	1.7686	+0.2%	11 → 12
T2-F1	122	2.0610	2.0483	-0.6%	12 → 13
T2-F2	144	1.2368	1.2448	+0.6%	1 → 1
T2-F3	121	1.4984	1.4980	-0.0%	2 → 2
T2-F4	167	1.1736	1.1690	-0.4%	2 → 2
127+	122	2.0610	2.0483	-0.6%	12 → 13
1–21	137	1.1258	1.1219	-0.4%	1 → 1
22–63	158	1.3111	1.3135	+0.2%	1 → 1
64–126	137	1.4161	1.4198	+0.3%	3 → 3
joint	236	1.7469	1.7382	-0.5%	13 → 14
single	318	1.2407	1.2434	+0.2%	4 → 4

All smaller groups are shown, even when below the veto threshold. Full CV and practice group tables are in ablation-breakdowns.csv.

Capped cards and known weaknesses

0 removed, 1 new capped evaluation rows across historical and practice. Overlapping practice/historical outcomes are separate rows, not independent events.

Partition	Change	Card suffix / unit	Assets	As-of	v1 → v1.1
held-out	new	01446	UST_2Y,UST_10Y	2022-02-11	7.994 → 8.000

Full card identifiers and horizon labels: capped-changes.csv.

Previously identified 2022 regressions

Historical subset	Cards	v1 mean	v1.1 mean	Caps
Monthly macro, 2022	8	2.7957	2.6590	1 → 1
Rates, 2022	55	1.8842	1.8908	1 → 2
All historical, 2022	119	1.7295	1.7288	4 → 5

The flagged CPI card synthetic-…-00895 stays capped: 8.000 → 8.000. The frozen winner keeps the original monthly trend and 25% trailing drift. Monthly damping and the 36-change window were tested, but worsened practice mean and combined cap counts. Momentum-sign drift reduced some practice losses but failed the CV tradeoff. None can be retained under C1.

These are post-selection diagnostics. Late-2021 zero-bound rates, 2008 and COVID transitions remain important stress cases. The monthly series use revised historical vintages; improvements here do not establish real-time macro accuracy.

Reproducibility and delivery

Frozen rules

```
{
  "drift": 0.25,
  "volatility": "recent",
  "tails": "fhs5",
  "monthly": true,
  "joint": true,
  "width": 1.0
}
```

Only the tail law changes from numeric_v1. FHS uses five-change blocks and a lagged 60-change half-life filter. The immutable Phase 2 numeric_config.json and frozen-selection.json are unchanged. New configuration: reports/phase2_5/numeric_v1_1_config.json; runtime copy: t2agent/numeric_v1_1_config.json. The protocol was committed as da26396 before scores. All 55,005 archived evaluation files matched their published manifest.

Operational validation

133 tests pass; pip check passes. 104/104 official admissibility checks pass at 2,000 draws (maximum host forecast time 0.072 s). All three linux/amd64 images pass the offline smoke test with read-only root. The full official scorer matches all 2,060 historical scores within 8.9e-16 and all 88 practice scores within 2.3e-16; zero inference failures. The ten-asset output triplet is 418,902 bytes, well below 64 MiB.

CI passed 133 tests, 104/104 admissibility gates, and the amd64 offline smoke test. The Public image was pulled with an empty Docker credential config and that exact digest passed g0–g3 offline with no fallback. Successful CI.

Immutable image: `ghcr.io/hhwolf/agenthon-numeric-v1@sha256:f3ec5167b28f0172004a78e336d84a4ecd423d27da42d942693e51cfc085c507`

Convenience tag: `numeric_v1_1`. CI tag: `numeric-v1.1-20261015-1`. Evidence: publication.json, anonymous-pull.json and published-image-smoke.json.

Limits and reproduction

CV and practice were used for selection. Year-block CIs do not remove that selection bias. Practice includes 85 inferred business-day target sets and three explicit monthly sets. Macro values include revisions; EM shapes can use G10 substitutes; historical calendars are approximations. No sealed outcomes or organizer feedback were used. No competition pack or submission upload was run.

Restore data per reports/EVALUATION_DATA.md. Run `.venv/bin/python -m t2eval.phase2_5` in a fresh output copy; it refuses to overwrite a freeze. Then run `make test`, `make admissibility FORECASTER=numeric_v1_1`, and `make backtest FORECASTER=numeric_v1_1`. Every ablation calls the pinned official scoring implementation.